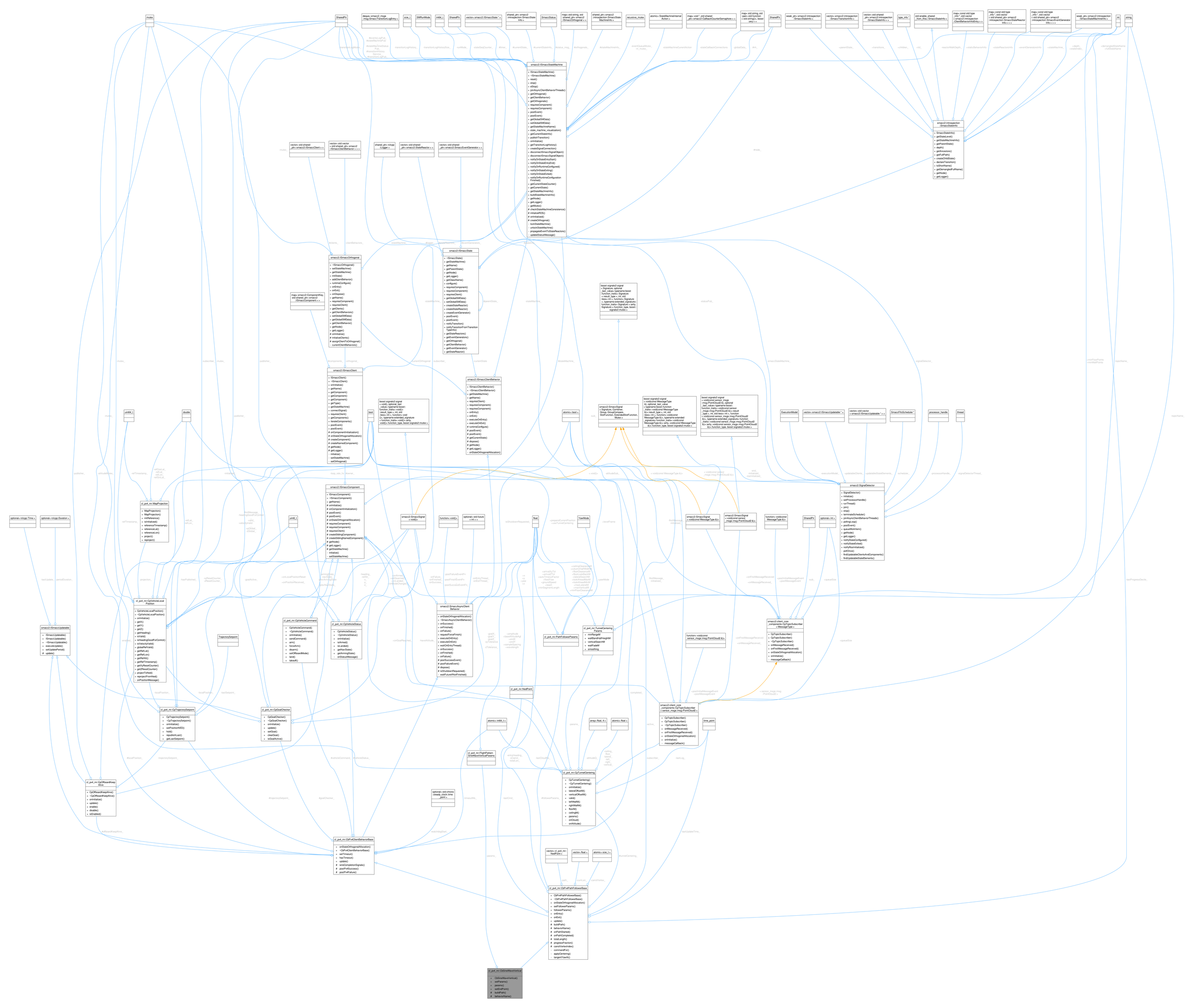

mutex
SharedPtr
deque< smacc2_msgs ::msg::SmaccTransitionLogEntry >
size_t
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2 ::introspection::SmaccStateInfo >
SmaccStatus
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
recursive_mutex
atomic< StateMachineInternal Action >
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, bool> ,any* > >
SharedPtr
weak_ptr< smacc2::introspection ::SmaccStateInfo >
vector< smacc2::introspection ::SmaccTransitionInfo* >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo* > >
type_info *
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info* > > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info* > > >
weak_ptr< smacc2::introspection ::SmaccStateMachineInfo* >
int
string
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
checkStateMachineConsistance()
initialiseROS()
onInitialized()
createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
vector< std::shared _ptr< smacc2::ISmaccClient > >
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::ISmaccStateReactor > >
vector< std::shared _ptr< smacc2::ISmaccEventGenerator > >
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
smacc2::ISmaccOrthogonal
+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
onInitialize()
initializeClients()
assignClientToOrthogonal()
- currentClientBehaviors()
smacc2::ISmaccState
+ ~ISmaccState()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvents()
onComponentInitialization()
onStateOrthogonalAllocation()
createComponent()
createNamedComponent()
getNode()
getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
runtimeConfigure()
postEvent()
postEvent()
getCurrentState()
dispose()
getNode()
getLogger()
- onStateOrthogonalAllocation()
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended _signature< function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature > ::result_type >, int, std ::less< int >, function< Signature >, typename extended _signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunctionType, Extended SlotFunction, Mutex >
boost::signals2::signal < void(const MessageType &), optional_last_value < typename boost::function _traits< void(const MessageType &)>::result_type, int, std ::less< int >, function< void(const MessageType &)>, typename extended _signature< function_traits< void(const MessageType &)>::arity, void(const MessageType &)>::function_type, boost::signals2::mutex >
boost::signals2::signal < void(const sensor_msgs ::msg::PointCloud2 &), optional_last_value < typename boost::function _traits< void(const sensor _msgs::msg::PointCloud2 &)>::result _type, int, std::less< int >, function < void(const sensor_msgs::msg::PointCloud2 &)>, typename extended _signature< function_traits< void(const sensor_msgs::msg::PointCloud2 &)>::arity, void(const sensor_msgs::msg::PointCloud2 &)>::function_type, boost::signals2::mutex >
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
SmaccFifoScheduler *
processor_handle
thread
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
uint64_t
double
bool
atomic< bool >
optional< rclcpp::Time >
optional< rclcpp::Duration >
cl_px4::MapProjection
+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()
uint8_t
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
onInitialize()
onComponentInitialization()
postEvent()
postEvent()
onStateOrthogonalAllocation()
requiresComponent()
requiresComponent()
requiresClient()
createSiblingComponent()
createSiblingNamedComponent()
getNode()
getLogger()
getStateMachine()
- initialize()
- setStateMachine()
smacc2::SmaccSignal < void() >
function< void()>
optional< std::future < void > >
float
YawMode
smacc2::SmaccSignal < void(const MessageType &) >
smacc2::SmaccSignal < void(const sensor _msgs::msg::PointCloud2 &) >
function< void(const MessageType &)>
SharedPtr
optional< int >
cl_px4::CpVehicleLocal Position
+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ execute(Update)
+ setUpdatePeriod()
update()
cl_px4::CpVehicleCommand
+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()
cl_px4::CpVehicleStatus
+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
postSuccessEvent()
postFailureEvent()
dispose()
isShutdownRequested()
- waitFutureIfNotFinished()
cl_px4::PathFollowerParams
cl_px4::TunnelCentering Params
+ minRangeM
+ wallBackHalfHeightM
+ verticalSearchM
+ wallFadeM
+ smoothing
smacc2::client_core _components::QpTopicSubscriber < MessageType >
+ QpTopicSubscriber()
+ QpTopicSubscriber()
+ ~QpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
function< void(const sensor_msgs::msg::PointCloud2 &)>
smacc2::client_core _components::QpTopicSubscriber < sensor_msgs::msg::PointCloud2 >
+ QpTopicSubscriber()
+ QpTopicSubscriber()
+ ~QpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
time_point
cl_px4::NedPoint
TrajectorySetpoint
cl_px4::QpTrajectorySetpoint
+ QpTrajectorySetpoint()
+ ~QpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ requestAndHoldLast()
+ getLastSetpoint()
cl_px4::QpGoalChecker
+ QpGoalChecker()
+ ~QpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()
atomic< int64_t >
array< float, 4 >
atomic< float >
cl_px4::FlightPattern SineWaveVerticalParams
cl_px4::CpTunnelCentering
+ CpTunnelCentering()
+ ~CpTunnelCentering()
+ onInitialize()
+ lateralOffsetM()
+ verticalOffsetM()
+ valid()
+ leftWallM()
+ rightWallM()
+ floorM()
+ ceilingM()
+ params()
+ onCloud()
- onAttitude()
optional< std::chrono ::steady_clock::time _point >
cl_px4::QpPx4ClientBehaviorBase
+ onStateOrthogonalAllocation()
+ ~QpPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
wireCompletionSignals()
postPx4Success()
postPx4Failure()
vector< cl_px4::mr::NedPoint >
vector< float >
atomic< size_t >
cl_px4::QbPx4PathFollowerBase
+ QbPx4PathFollowerBase()
+ ~QbPx4PathFollowerBase()
+ onStateOrthogonalAllocation()
+ setFollowerParams()
+ followerParams()
+ onEntry()
+ onExit()
+ update()
buildPath()
behaviorName()
onPathStarted()
onPathCompleted()
totalLength()
progressFraction()
carrotVertexIndex()
- commandFor()
- applyCentering()
- tangentYawAt()
cl_px4::mr::QbSineWaveVertical
+ QbSineWaveVertical()
+ setParams()
+ params()
+ setEndPoint()
buildPath()
behaviorName()